



NX-421: Neural Signals and Signal Processing

Miniproject 1

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Contributions

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Gianmarco Placì	414573	Skull Stripping, Concatenation, Motion correction, K-means
Vincenzo Riviaccio	409465	Coregistration, Smoothing, Design matrix
Pasquale Scognamiglio	416558	Intro, K-means

Table 1: Contributions

1 Introduction

This miniproject explores the neural correlates of motor activity using functional magnetic resonance imaging (fMRI), aiming to identify and compare the brain regions that are activated during hand and foot movements. The used dataset originates from the Human Connectome Project [1], and corresponds to a motor task designed to map cortical motor areas and adapted from by Buckner and colleagues [2].

2 Materials and Methods

2.1 Data

Data used for this miniproject consists of fMRI recordings from a single patient who was instructed to perform finger tapping in both hands, toes squeezing for both feet, and tongue motion. The experiment was carried in a block design, with every block containing approximately 12 seconds of active movement, preceded by a visual cue lasting around 3 seconds and fixation periods. All blocks came with time and labels carrying information about onset, duration, and type of each task condition. In addition to the functional recordings, a high-resolution T1-weighted (T1w) anatomical image was used for spatial registration and anatomical reference.

2.2 Preprocessing Steps

2.2.1 Structural

Skull Stripping Structural preprocessing involved at first skull stripping on the T1-weighted anatomical image to separate the brain tissue from non-brain structures, removing skull, scalp and other surrounding tissues (Figure 1 and 2).

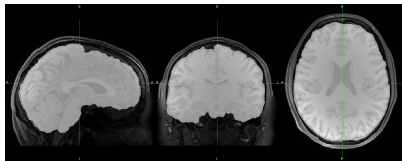


Figure 1: T1w with mask.

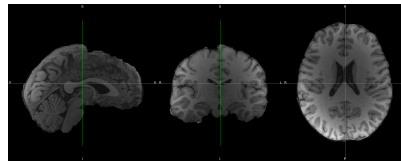


Figure 2: After skull stripping.

Segmentation Tissue segmentation was performed after brain extraction to classify the voxels into gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF) within the brain-only configuration (Figure 4).

2.2.2 Functional

Concatenation The two functional runs were first normalized before concatenation for comparability. A brain mask was obtained by skull-stripping the temporal mean of the first run and resampled to the second run's space. Each run was then treated setting the global variance of brain voxel intensities to one, ensuring comparable signal scales across runs.

Motion correction Head-motion correction was applied independently to each run using a six-degree-of-freedom rigid-body transformation estimated relative to the mean functional image to account for movement artifacts. Framewise displacement (FD) was computed from the estimated motion parameters following methodology proposed by Power et al. [3]. The FD traces for the two runs are shown in Figure 3, with high-motion volumes that were identified using an interquartile range (IQR)-based threshold. The motion-corrected runs were then concatenated in time, yielding a single preprocessed functional dataset.

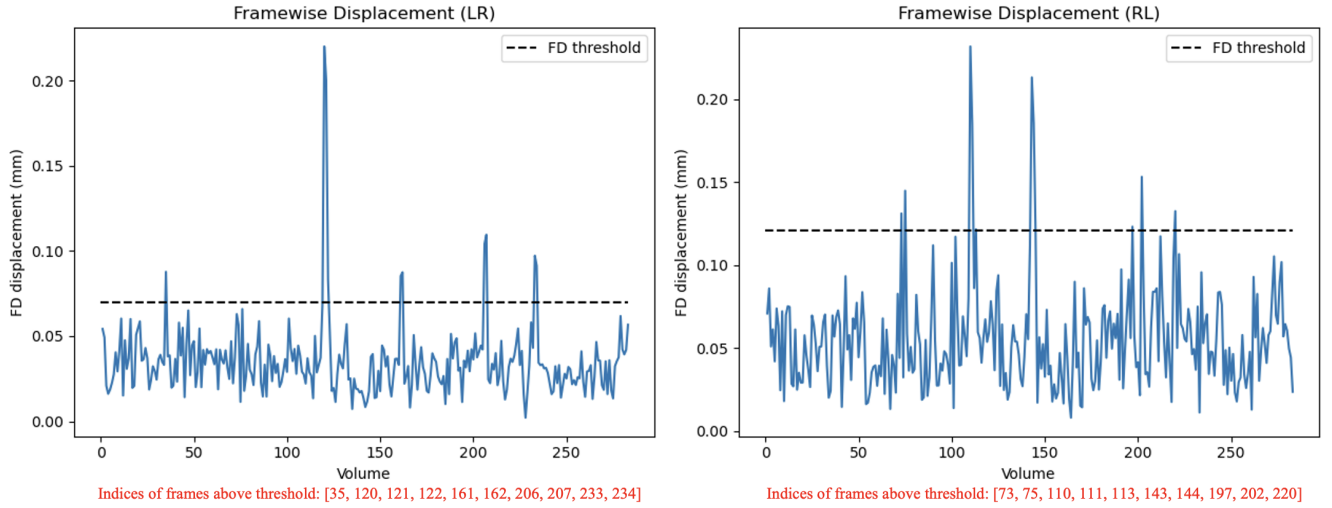


Figure 3: Framewise displacement for LR and RL runs with marked suboptimal frames.

Coregistration Functional-anatomical coregistration was performed to map activation estimates on the corresponding anatomical locations. A six degree of freedom affine transformation was estimated, optimizing boundary-based alignment between functional and structural images. The result is shown in Figure 5.

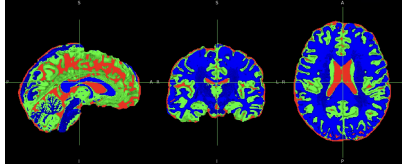


Figure 4: Brain tissue segmented.

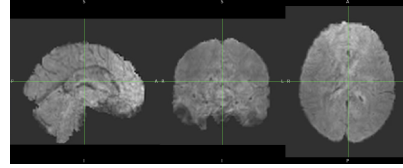


Figure 5: Functional-structural coreg.

Spatial smoothing Finally, spatial smoothing was applied to the concatenated dataset using a Gaussian kernel with a full width at half maximum (FWHM) of 6 mm (corresponding to $\sigma = 6/2.3548 \approx 2.55$ mm), reducing high-frequency spatial noise. Smoothing increased SNR by averaging neighboring voxels and helped meeting the Gaussian assumptions of the GLM used for statistical purposes (Figure 6 and 7).

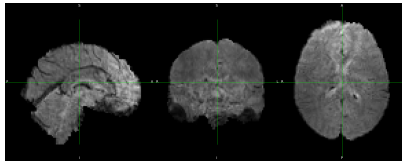


Figure 6: fMRI before smoothing.

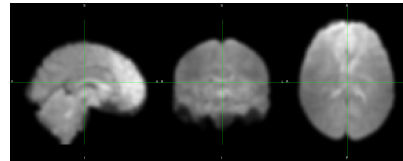


Figure 7: fMRI after smoothing.

2.3 General Linear Model (GLM) Analysis

Experimental Design Matrix Based on the task event file provided for the subject, the experimental design matrix was constructed. All of the experimental conditions (left hand, right hand, left foot, right foot, tongue, fixation and cue) were modeled as a separate regressor, together with third-degree drift regressors to account for low frequency oscillations (Figure 8).

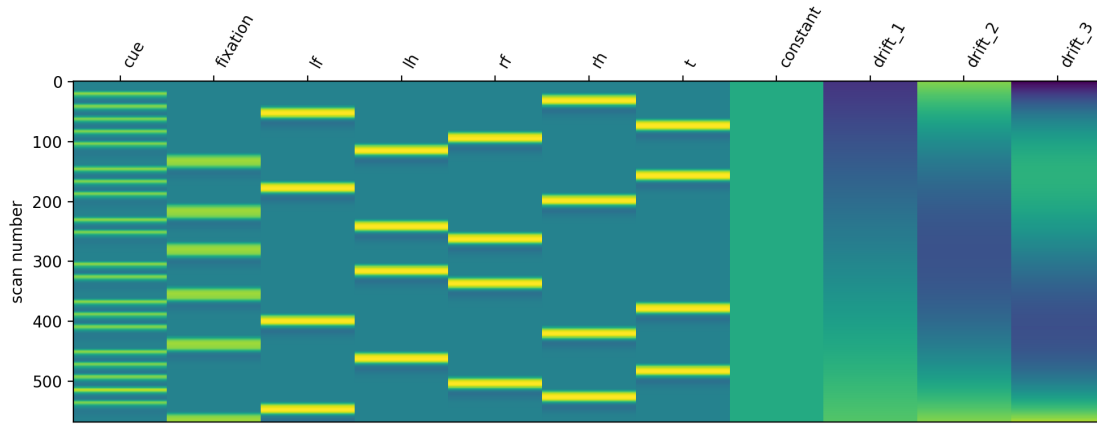


Figure 8: Design matrix of the blocks of the experimental setup.

First-Level GLM Estimation A voxel-wise GLM was fitted to the functional data using an AR(1) noise model, a 0.01 Hz high-pass filter, and the SPM canonical HRF. A brain mask restricted the analysis to gray matter voxels. For each experimental condition, voxel-wise β - and Z-maps were estimated, representing respectively the amplitude of the BOLD response and its statistical significance relative to the baseline ($Z > 3$ for visualization). The Z-maps in Figure 9 emphasize voxels where these task-related activations are statistically significant. For example, when the cue appears, it's possible to see higher activation of the occipital area, where visual information is processed. It's also possible to observe that in the tasks related to the right part of the body there's an activation of the motor area in the left hemisphere (and viceversa), with a more specific distinction between the foot and hand on the same side depending on their position in the motor homunculus.

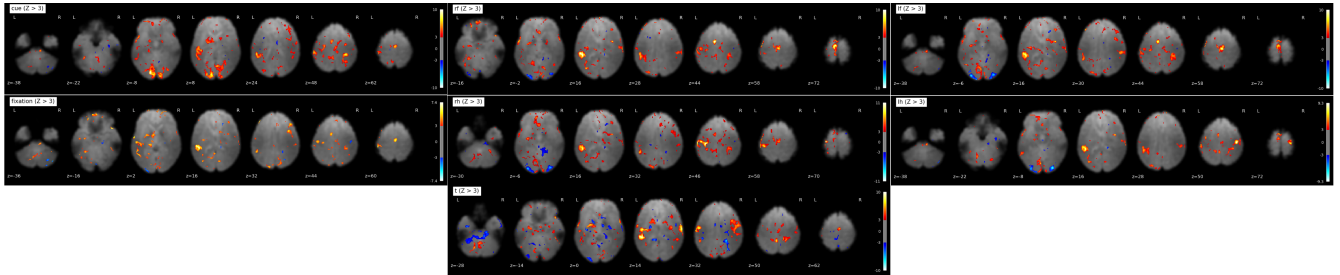


Figure 9: Z-maps for all conditions ($Z > 3$, $p < 0.001$).

Contrast: Hand vs. Foot Movement To investigate differential activation patterns between hand and foot movements, a contrast vector was defined as $c = [0, 0, +1, +1, -1, -1, 0, 0, 0, 0, 0]$ corresponding to $(LH + RH) - (LF + RF)$. This contrast isolated regions that were more active during hand movements relative to foot movements. The resulting contrast map was computed and is shown in Figure 10.

Anatomical Localization with the AAL Atlas The Hand > Foot Z-map was overlaid with the AAL (SPM12) atlas to anatomically localize the activation peaks (Figure 10). The atlas was resampled to match the functional resolution, and mean Z-scores were computed within each labeled region. The regions showing the highest mean positive Z-values are listed in Table 2, revealing strongest activation in the pre and post central gyri, consistent with the motor and somatotopic hand representation in the cortex.

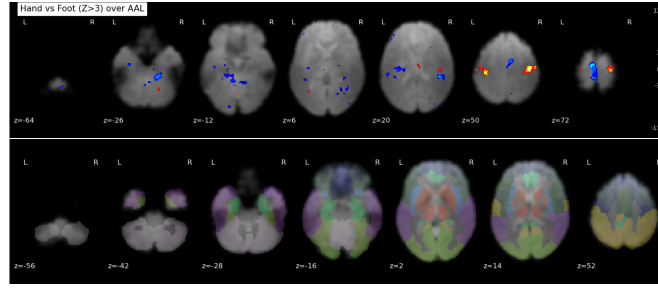


Figure 10: Above the hand vs foot contrast map ($Z > 3$, $p < 0.001$). Below the AAL atlas overlay used for the mean Z computation reported in Table 2.

2.4 Theoretical: Part 1

So far, we have worked on data from a single subject and extracted activation maps for the hand-foot contrast. This analysis was performed at the first within subject level, producing a single contrast map that captures task-related activation for one individual. Since the current dataset includes only one subject, a true second-level analysis can't be performed, as it requires multiple independent subjects to estimate variability between them and enable statistical inference at the group level.

If we had 10 subjects of different known ages performing the same motor tasks, then a second-level analysis would become feasible. In this case, each subject's first-level contrast map would become an observation in another general linear model. A design matrix could for example consist of an intercept and an age covariate, with relevant vectors being $[1, 0]$ to test the overall mean activation, and $[0, 1]$ to test for an age-related modulation of activity. These would allow answering questions such as whether hand vs foot activation is consistent across subjects or is modulated by age.

2.5 Variant 3: K-means clustering

K-means data driven clustering was used to identify dominant patterns in the fMRI data and to represent characteristic brain states recurring in time. Each volume was considered as a sample for the analysis, with the voxels being features. The algorithm performed a grouping of such samples by minimizing the Within-Cluster Sum of Squares (WCSS). The elbow method showed a plateau in the reduction of the described within-cluster variance with only 2 clusters, but for coherence to the experimental task, 7 clusters were used and the axial brain maps corresponding to the resulting centroids reported (Figure 12). Pearson correlation was then calculated for the first 5 centroids, and results are reported in Figure 11.

Table 2: Top AAL regions for the Hand > Foot contrast (mean Z within label).

AAL Region	Mean Z
Precentral_R	0.77
Postcentral_L	0.65
Postcentral_R	0.61
Precentral_L	0.56
Cerebellum_6_L	0.53
Frontal_Sup_Medial_R	0.43
Vermis_6	0.41
Cerebellum_6_R	0.40
Cerebellum_4_5_L	0.38
Frontal_Sup_Medial_L	0.38

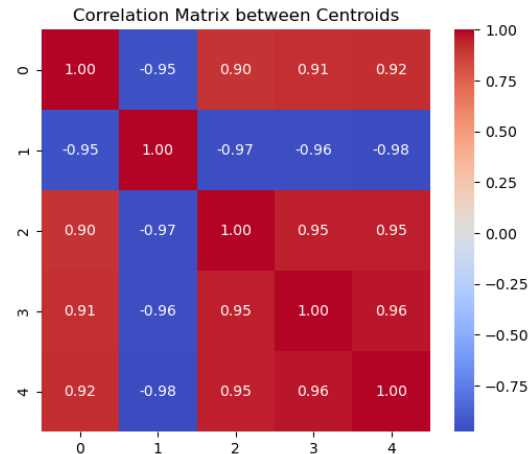


Figure 11: Similarity matrix among 5 centroids.

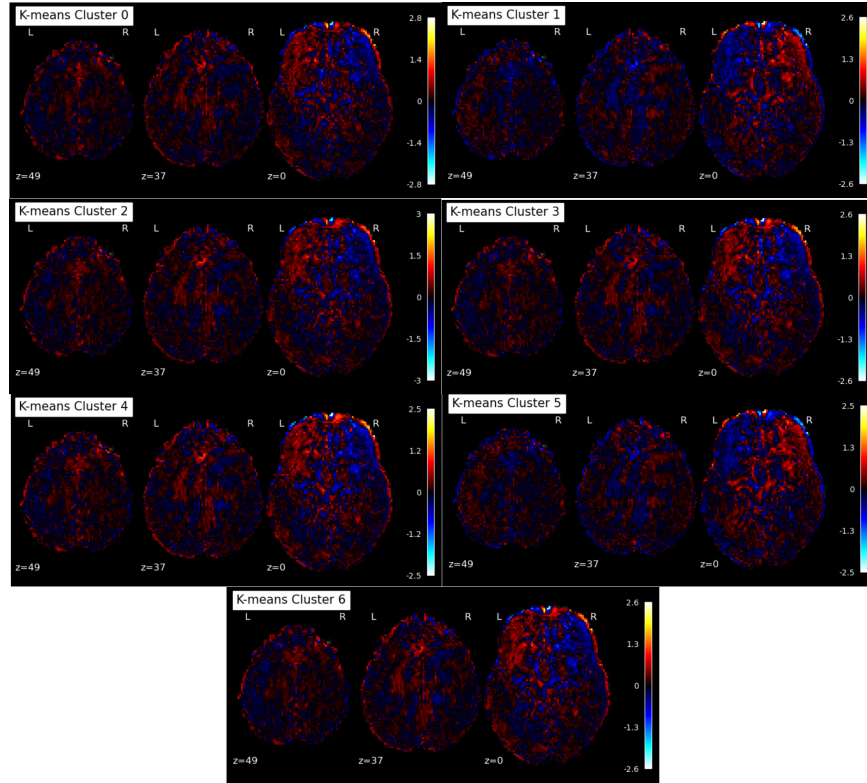


Figure 12: Axial brain map for 7 centroids

2.6 Theoretical: Part 2

Brain activity is organized into functional networks that have been characterized in connectivity studies [4]. An inspection of the obtained brain activation patterns reveals correspondence to some of these identified networks. In particular, increased activation is observed within the primary motor and somatosensory cortices, consistent with the engagement of the localized sensorimotor network. Other interesting observations refer to the activity recorded in the occipital cortex, which likely indicates recruitment of the visual network, as visual stimuli preceded each movement block with cues. Finally, the engagement of the dorsal attention and frontoparietal control networks can also be inferred from activity in parietal and frontal regions, reflecting the top-down attentional control and stimulus-response selection, fundamental for cue to action transition.

As for the methods used for the analysis of activation patterns, the GLM allowed voxel-wise statistical testing of task-related effects by explicitly modeling the experimental design and finding regions where the BOLD signal significantly varies across conditions. For the aim of this project, it enabled the localization of distinct motor areas associated with the different conditions. On the other hand, the multivariate pattern analysis considered distributed activation patterns across volumes to uncover large-scale functional networks and determine whether task-related information could be seen in neural activity, without testing the changes in the signal, but rather finding different activation patterns in the data with no labels. Such method would have been more suitable if investigating representational content within an already localized area, whereas the GLM performed better in terms of large-scale functional topography mapping thanks to its hypothesis driven structure.

References

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